Understanding Token Specialization in Distilled Vision Transformers
under Severe Data Imbalance

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Abstract

Vision Transformers (ViTs) are highly effective at scale but exhibit severe performance degradation on long-tailed data distributions. Recent approaches successfully mitigate this by leveraging Knowledge Distillation (KD), employing a specialized Distillation (`DIST`) token to transfer robust representations from a Convolutional Neural Network teacher to a ViT student. During inference, current methodologies rely on a rigid heuristic: a static 50/50 averaging of the Class (`CLS`) and `DIST` token predictions. While this achieves strong baseline performance, the internal behavior of these tokens under data starvation remains largely unexplored. In this paper, we conduct a structural audit of distilled ViTs and uncover empirical evidence of Token Specialization. Under severe data imbalance, the tokens do not learn redundant features; rather, they bifurcate. The `CLS` token saturates as a Head-class expert, while the `DIST` token specializes on Tail classes. Consequently, we demonstrate that fixed 50/50 fusion actively suppresses model performance by causing expert sabotage. To expose this, we introduce the Neural Entropy Router, a diagnostic mechanism that dynamically fuses tokens based on instance-level uncertainty, successfully recovering latent performance without architectural modifications. Finally, through a No-KD causal ablation, we provide strong evidence that Knowledge Distillation—rather than inherent architectural bias—drives this structural bifurcation. These findings suggest that dynamic token routing is crucial for fully leveraging distilled representations in imbalanced environments.

1 Introduction

Real-world datasets naturally exhibit long-tailed distributions, where a few majority (Head) classes contain the vast majority of samples, while numerous minority (Tail) classes suffer from severe data starvation. Vision Transformers (ViTs) lack the inductive biases of Convolutional Neural Networks (CNNs) and are notoriously data-hungry, making them highly susceptible to overfitting in these regimes.

To address this, state-of-the-art methods like DeiT-LT apply Knowledge Distillation (KD) to transfer robust feature representations from a pre-trained CNN teacher to a ViT student. This process introduces a specialized Distillation (`DIST`) token alongside the standard Class (`CLS`) token. During inference, the final prediction is computed by blindly averaging the logits of both tokens (a 50/50 split). While this engineering heuristic achieves high top-1 accuracy, it leaves a fundamental scientific question unexplored: *What actually happens inside the `CLS` and `DIST` tokens under severe data starvation?*

Rather than proposing a new transformer block, this work seeks to understand the existing ones. We investigate the structural behavior of these tokens and hypothesize that under long-tailed distributions, they bifurcate into highly specialized experts. Our contributions are as follows:

1. **Identifying Token Specialization:** We provide empirical evidence that the `CLS` token functions as a Head expert, while the `DIST` token specializes on

Tail classes.

2. **Exposing Expert Sabotage:** We show that fixed 50/50 fusion actively suppresses performance by forcing experts to dilute each other's confident predictions.

3. **Entropy-Guided Routing:** We propose dynamic, instance-level routing based on Shannon entropy as a diagnostic tool to recover this suppressed performance.

4. **Isolating the Causal Factor:** Through rigorous ablation, we isolate Knowledge Distillation as the primary causal factor driving token bifurcation.

2 Related Work

Vision Transformers and Distillation Data-efficient Image Transformers (DeiT) introduced a token-based distillation strategy, where a CNN teacher supervises a dedicated `DIST` token. This allows the ViT to implicitly learn convolutional inductive biases without altering the self-attention mechanism.

Long-Tailed Learning Standard approaches to long-tailed learning adjust the loss landscape. Deferred Reweighting (DRW) trains the network normally for early epochs to learn robust representations, before applying class-balanced loss weighting in later epochs.

Dynamic Routing Mixture of Experts (MoE) architectures dynamically route inputs to specialized subnetworks. However, these routers are integrated deep within the architectural blocks. In contrast, our approach operates entirely at the logit level post-hoc.

Calibration and Uncertainty Understanding model confidence is critical for routing. Post-hoc calibration methods often utilize Shannon entropy to measure predictive uncertainty. We leverage this principle to guide token fusion.

3 Methodology

Our methodology follows a scientific progression to investigate the token behavior.

3.1 Observation: The Bifurcation

We trained a DeiT-Tiny architecture on CIFAR-10-LT (Imbalance Factor 50). Analyzing the class-wise confusion matrices revealed a striking structural bifurcation. The `CLS` token exhibited diagonal saturation on the most frequent Head classes but bled heavily across the Tail. Conversely, the `DIST` token demonstrated intense diagonal saturation on the rarest Tail classes.

3.2 Hypothesis: Expert Sabotage

If the tokens are mutually exclusive experts, then a fixed fusion weight ($\alpha = 0.5$) where $L_{fused} = \alpha L_{cls} + (1 - \alpha)L_{dist}$ is fundamentally flawed. When predicting a Tail class, the `CLS` token (lacking tail representations) outputs high entropy noise, actively dragging down the `DIST` token's correct prediction.

3.3 Evidence: The Oracle Alpha Search

To prove the existence of an optimal routing policy, we conducted an empirical Oracle search, sweeping α from 0.0 to 1.0 for every sample. The Oracle revealed a step-function: Head classes required $\alpha \approx 0.80$, while Tail classes required $\alpha \approx 0.05$. The Oracle bound reached **74.40%** (vs the 72.10% baseline), confirming that dynamic routing could recover significant performance.

3.4 Contradiction and New Hypothesis

While class frequencies (n_c) dictate the optimal average α, class frequencies are unknown during true inference. By training a Random Forest on the Oracle's success cases, we discovered that class frequency alone was insufficient for routing. However, *DIST Entropy* accounted for 77.5% of the predictive importance. We hypothesized that instance-level uncertainty (entropy) could serve as a class-agnostic proxy for routing.

3.5 The Neural Entropy Router

We designed a lightweight Multi-Layer Perceptron (MLP) to dynamically predict α based on the output distributions:

$$X = [\max(P_{cls}), \max(P_{dist}), H(P_{cls}), H(P_{dist})] \quad (1)$$

where H is the Shannon entropy. The router learns to trust the token with the lowest uncertainty for any given instance.

4 Experiments

All experiments were conducted utilizing PyTorch with Automatic Mixed Precision. Models were trained for 300 epochs with a batch size of 256.

4.1 Baseline, Oracle, and Router

At Imbalance Factor 50, the 50/50 fusion achieved 72.10%. By applying the Neural Entropy Router, we achieved **73.73%**, successfully recovering a large portion of the Oracle gap. The Router autonomously recognized the structural superiority of the `DIST` token, routing the majority of instances toward it ($\alpha \approx 0.05$).

4.2 Seed Variance and Imbalance Scaling

Changing the global random seed yielded nearly identical router performance (74.74%), confirming this is a structural reality, not a statistical fluke.

We then scaled dataset severity:

- **IF=100 (Extreme):** The gap widened. `DIST` (69.5%) outperformed `CLS` (63.8
- **IF=10 (Mild):** The dataset was healthier. `CLS` (81.0%) caught up to `DIST` (81.5%). The router synergistically blended them ($\alpha \approx 0.30$) to achieve **82.17%**.

4.3 Architecture Scaling

To test scale-invariance, we evaluated DeiT-Small (22M params) and DeiT-Base (86M params). While larger models overfit the small dataset, the structural dynamic held perfectly: the `DIST` token outperformed the `CLS` token by exactly ∼**4.6%** across Tiny, Small, and Base.

4.4 The Causal Test: No-KD

To isolate Knowledge Distillation as the cause, we trained the exact same architecture on CIFAR-10-LT, but *removed the Teacher model*. The tokens collapsed symmetrically, achieving identical Tail accuracies (65.8% vs 66.5%). Consequently, the Neural Router converged to a random blending weight ($\alpha = 0.249$) and achieved zero improvement over the baseline. This provides strong empirical evidence that KD drives Token Specialization.

5 Discussion

Why does DIST dominate the Tail? We hypothesize that this specialization stems from differing inductive biases. The CNN teacher possesses strong translational equivariance, making it inherently more robust to data starvation. The distillation loss anchors the `DIST` token to this robust, dense knowledge. Conversely, the `CLS` token (guided only by standard cross-entropy) suffers from gradient starvation on the Tail and rapidly overfits to the Head classes.

Limitations and Failure Cases Our routing mechanism operates purely at the logit level during inference. While effective, it relies on entropy heuristics, which can be miscalibrated on out-of-distribution (OOD) data. Furthermore, while the Router significantly improves over the baseline, it does not fully close the Oracle gap, suggesting that logit-level features alone are insufficient for perfect token disambiguation.

Future Work Future research should investigate whether this Token Specialization can be explicitly encouraged during the representation learning phase, rather than merely managed post-hoc. Expanding this causal audit to massive long-tailed datasets, such as ImageNet-LT, remains a critical next step to validate the universality of these findings.

6 Conclusion

In this work, we audited the internal dynamics of distilled Vision Transformers under severe data imbalance. We provided strong empirical evidence that Knowledge Distillation causes the `CLS` and `DIST` tokens to bifurcate into highly specialized experts. By exposing the flaw in fixed 50/50 averaging and introducing an entropy-guided diagnostic router, we demonstrated that dynamic, instance-level routing is necessary to unleash the true potential of the Knowledge Distillation token.